

3(a)	1 $(q^2 = \frac{440}{1100} \text{ or } q^2 = 0.4)$ $q = \sqrt{0.4}$ or $q = 0.632455532$; 2 $(p = 1 - 0.632455532 \text{ or } p = 0.367544468)$ $2pq = 2 \times 0.632455532 \times 0.367544468$ or $2pq = 0.4649110641$; 3 (heterozygotes = $2pq \times 1100$ e.g. 511.4 or 511.5 or 511.6672) 511 / 512 ;	3
3(b)	any two from: 1 colours / shades / categories / phenotypes, not distinct / not discrete / overlap / form a (gradual) range / include intermediates ; 2 >1 / 4 / several / many, genes ; 3 environment affects, it / colour / phenotype ;	2
3(c)(i)	any two from: 1 (as climate changes) increase in CO₂ concentration causes, stomatal aperture to decrease / stomata to be less open ; 2 less water, lost / evaporated / transpired (through stomata) ; 3 compensates for / allows survival in, low water availability / dry conditions ;	2

Question	Answer	Marks
3(c)(ii)	any four from: 1 (abscisic acid / ABA) binds to receptor on, (guard) cell surface membrane ; 2 H^+ stops moving, out of cell / into cell wall ; 3 Ca^{2+} ions enter, cytoplasm / cell or 4 Ca^{2+} ions act as second messenger ; 4 K^+ ions leave cell ; 5 water leaves cell, by osmosis / down water potential gradient ; 6 less turgid / flaccid, guard cell(s) close stoma(ta) ;	4
3(d)(i)	any two from: 1 0.1, (μmol) increases / gives higher, rate (compared to control / normal) ; 2 1(.0) and 10(.0), (μmol) decrease / give lower, rate (compared to control / normal) ; 3 10(.0) (μmol) rate, stays (approx.) constant / plateaus / does not increase, over, time / (6) days ; 4 data quote comparing rate at two concentrations on one day or data quote comparing rate on two different days for 10 μmol ;	3
3(d)(ii)	1 protons / H^+ (ions) ; 2 cellulose ;	2

Question	Answer	Marks
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